But disorder of what? As we have seen, black holes appear to be terribly simple objects, so what is the source of this overwhelming disorder? On this question, Hawking's calculations were completely silent. His partial merger of general relativity and quantum mechanics could be used to find the numerical value of a black hole's entropy, but offered no insight into its microscopic meaning. For nearly a quarter of a century, some of the greatest physicists tried to understand what possible microscopic properties of black holes could account for their entropy. But without a fully trustworthy amalgam of quantum mechanics and general relativity, glimpses of an answer may have been uncovered, but the mystery remained unsolved.

Enter String Theory

Or, it did until January 1996, when Strominger and Vafa—building on earlier insights of Susskind and Sen—released a paper to the electronic physics archive entitled "Microscopic Origin of the Beckenstein-Hawking Entropy." In this work, Strominger and Vafa were able to use string theory to identify the microscopic constituents of a certain class of black holes and to calculate precisely their associated entropy. Their work relied on the newfound ability to partially circumvent the perturbative approximations in use during the 1980s and early 1990s, and the result they found agreed exactly with that predicted by Bekenstein and Hawking, finally completing a picture partially painted more than two decades previously.

Strominger and Vafa focused on the class of so-called extremal black holes. These are black holes that are imbued with charge—you can think of it as electric charge—and that, moreover, have the minimal possible mass consistent with the charge they carry. As can be seen from this definition, they are closely related to the BPS states discussed in Chapter 12. In fact, Strominger and Vafa exploited this similarity to the hilt. They showed that they could build—theoretically, of course—certain extremal black holes by starting with a particular collection of BPS branes (of certain specified dimensions) and binding them together according to a precise mathematical blueprint. In much the same way that an atom can be built—theoretically, again—by starting with a bunch of quarks and electrons and then precisely arranging them into protons and neutrons, surrounded by orbiting electrons, Strominger and Vafa showed how some of the newfound